

ICT600 CORRECTED STUDENT SCHEME

December 2019

Question-ordered model answers and marking guide

Scope	December 2019 final examination
Arrangement	Answers follow the original paper from Question 1 through Part B. Coding, diagrams, and written responses are integrated once at their correct positions.
Assessment	100 marks: Part A 70, Part B 30
Status	Corrected student reference; equivalent technically correct answers may receive credit.

How to use this scheme

Question-order rule

Attempt the paper first. Then check the matching answer in this document. Each answer appears once, in the same order as the question paper.

- Equivalent correct wording, diagrams, algorithms, and code may receive full credit when they meet the stated requirements.
- For coding, award method marks when a minor syntax error does not invalidate an otherwise correct solution.
- For design or illustration questions, the labelled diagram is the model answer; prose alone is not treated as a complete design.

Part A

Question 1(a)

6 marks

Describe Web Application, Web Server, and Web Browser.

Mark allocation

Awardable element	Marks
Web application definition	2
Web server definition	2
Web browser definition	2
Total	6

Complete model answer

- Web application: software accessed through a browser over a network, with processing on the client, server, or both.
- Web server: software or a system that receives HTTP requests and returns static or generated responses.
- Web browser: client software that requests resources, renders HTML/CSS, and executes client-side JavaScript.

Question 1(b)

4 marks

List FOUR types of cloud services.

Mark allocation

Awardable element	Marks
SaaS	1
PaaS	1
IaaS	1
A fourth valid, consistently categorised service model	1
Total	4

Complete model answer

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)
- Function as a Service (FaaS/serverless)

Marker note

The current notes emphasise SaaS, PaaS, and IaaS. Accept another correctly identified cloud-service category for the fourth mark.

Question 2(a)**4 marks**

Explain TWO target-attribute values for HTML links.

Mark allocation

Awardable element	Marks
First valid target and effect	2
Second valid target and effect	2
Total	4

Complete model answer

- `_self` opens the linked document in the current browsing context.
- `_blank` opens it in a new tab or window, depending on browser settings.
- Also accept `_parent` or `_top` with an accurate explanation.

Question 2(b)**6 marks**

Name SIX components of a website.

Mark allocation

Awardable element	Marks
Containing block	1
Logo	1
Navigation	1
Content	1
Footer	1
Whitespace	1
Total	6

Complete model answer

- Containing block, logo, navigation, content, footer, and whitespace.

Question 3(a)

4 marks

State TWO JavaScript variable-naming rules.

Mark allocation

Awardable element	Marks
First valid rule and explanation	2
Second valid rule and explanation	2
Total	4

Complete model answer

- A name may start with a letter, underscore, or dollar sign, but not a digit.
- Names are case-sensitive and cannot be reserved keywords.

Question 3(b)

6 marks

Describe TWO ways to include jQuery in a web page.

Mark allocation

Awardable element	Marks
Local-file method and correct inclusion	3
CDN method and correct inclusion	3
Total	6

Complete model answer

- Download the library and reference the local file with a script element.
- Reference a trusted CDN URL with a script element.

Question 4(a) - PHP loop trace

6 marks

Produce the exact output of the supplied while and do-while code.

Mark allocation

Awardable element	Marks
First heading	1
Correctly shows no while-loop number	2
Second heading	1
Correct do-while value	1
Explains pre-test versus post-test behaviour	1
Total	6

Complete model answer

```
with while:
(no number is printed)
with do-while:
1
```

Question 4(b) - PHP function

6 marks

Write the complete function and calls that reproduce all three required lines.

Mark allocation

Awardable element	Marks
Function and two parameters	2
Correct concatenated output	2
Three correct calls	2
Total	6

Complete model answer

```
<?php
function showPerson(string $firstName, int $year): void {
    echo $firstName . ' Ahmed. Born in ' . $year . '<br>';
}
showPerson('Maslina', 1989);
showPerson('Hardy', 1990);
showPerson('Azmi', 1987);
?>
```

Marker note

The stem says surname, but the examples vary the first name and keep Ahmed fixed. Accept any coherent implementation that reproduces the three lines.

Question 4(c) - indexed array loop

8 marks

Use an indexed PHP array and display Axia, MyVi and Bezza.

Mark allocation

Awardable element	Marks
Correct indexed array	3
Correct loop	3
All three values displayed	2
Total	8

Complete model answer

```
<?php
$scars = ['Axia', 'MyVi', 'Bezza'];
foreach ($scars as $scar) {
    echo $scar . '<br>';
}
?>
```

Question 5(a) - Profil Pak Belang page

5 marks

Create the complete HTML page shown in Figure 1.

Mark allocation

Awardable element	Marks
Valid page and heading/paragraph structure	1
Required text formatting	1
pak-belang.png image	1
Complete two-column table	1
Source and content completeness	1
Total	5

Complete model answer

```
<!doctype html>
<html lang="ms">
<head><meta charset="utf-8"><title>Profil Pak Belang</title></head>
<body>
<article>
<h1>Profil Pak Belang</h1>
<p><strong>Harimau Malaya</strong> (<em>Panthera tigris jacksoni</em>)
    merupakan salah satu subspesies harimau belang...</p>

<p>Terdapat sembilan subspesies harimau belang...</p>
<table border="1">
<tr><th>Spesies</th><th>Nama Saintifik</th></tr>
<tr><td>Harimau Bali</td><td><em>Panthera tigris balica</em></td></tr>
<tr><td>Harimau Jawa</td><td><em>Panthera tigris sondaica</em></td></tr>
<tr><td>Harimau Kaspian</td><td><em>Panthera tigris virgata</em></td></tr>
</table>

<p>Source: <a href="https://harimau.my/kenali-harimau-malaya/">https://harimau.my/kenali-harimau-malaya/</a></p>
</article>
</body>
</html>
```

Question 5(b)

5 marks

Illustrate the web page rendered by the supplied seminar-registration HTML.

Mark allocation

Awardable element	Marks
Main heading and fieldset legend	1
Name and email controls	1
Gender and date controls	1
Institution and category controls	1
Register and cancel controls	1
Total	5

Complete model answer

- A correct rendered model is shown below. The answer should show the visible page, not repeat the source code.

Seminar Registration Form

Participant Registration Form

Name	
Email	
Gender: () Male () Female	
Date of birth	
Institution	
Category	Select category
<button>Register Me</button><button>Cancel</button>	

Question 5(c) - MYR to USD converter

10 marks

Create the full HTML and JavaScript converter at MYR 4.12 for USD 1.

Mark allocation

Awardable element	Marks
Input, label and conversion button	3
Event handler/function	2
Numeric validation	1
Correct MYR divided by 4.12	2
Formatted result	2
Total	10

Complete model answer

```
<!doctype html>
<html lang="en">
<body>
  <h1>Online Currency Converter from MYR to USD</h1>
  <label>MYR <input id="myr" type="number" min="0" step="0.01">/label>
  <button type="button" id="convert">Convert to USD</button>
  <script>
    document.getElementById('convert').addEventListener('click', () => {
      const myr = Number(document.getElementById('myr').value);
      if (!Number.isFinite(myr) || myr < 0) {
        alert('Enter a valid MYR amount.');
        return;
      }
      const usd = myr / 4.12;
      alert(`MYR ${myr.toFixed(2)} = USD ${usd.toFixed(2)}`);
    });
  </script>
</body>
</html>
```

Expected output / behaviour

MYR 5.00 = USD 1.21

Part B

Part B Question 1

8 marks

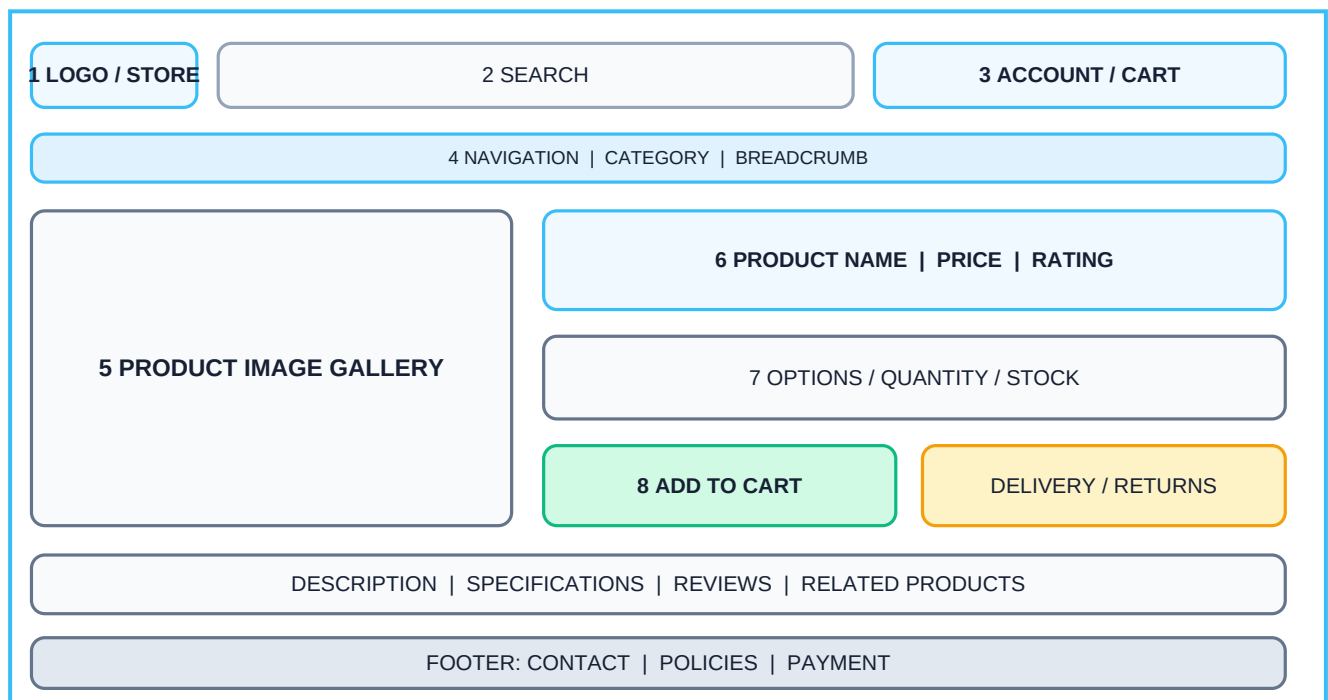
Design an e-commerce product-page wireframe containing EIGHT relevant features.

Mark allocation

Awardable element	Marks
Logo or store identity	1
Search	1
Account or cart	1
Navigation or breadcrumb	1
Product image gallery	1
Product name, price, or rating	1
Options, quantity, or stock	1
Add-to-cart action	1
Total	8

Complete model answer

- The labelled model wireframe below is one complete answer. Equivalent layouts receive full credit when the required features are visible and logically placed.



Part B Question 2 - PHP login and session flow

22 marks

Provide a complete runnable login solution matching the required guest, error and welcome states.

Mark allocation

Awardable element	Marks
Login form and email/password controls	5
Request and email validation	4
Credential decision and messages	5
Session start, default values and authenticated update	5
Safe output and coherent page flow	3
Total	22

Complete model answer

```
<?php
session_start();
$_SESSION['username'] ??= 'Guest';
$_SESSION['role'] ??= 'Guest';
$error = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $email = trim($_POST['email'] ?? '');
    $password = $_POST['password'] ?? '';
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        $error = 'Enter a valid email address.';
    } elseif ($email === 'amran@mydomain.com' && $password === 'Sec@1ret') {
        session_regenerate_id(true);
        $_SESSION['username'] = 'amran';
        $_SESSION['role'] = 'member';
    } else {
        $error = 'Incorrect username and/or password. Please login again.';
    }
}

function e(string $v): string { return htmlspecialchars($v, ENT_QUOTES, 'UTF-8'); }
?>

<!doctype html><html lang="en"><head><meta charset="utf-8"><title>Login</title></head>
<body>
<?php if ($_SESSION['role'] === 'member'): ?>
    <h1>Welcome</h1><p>Hello <?= e($_SESSION['username']) ?></p>
    <p>User role as <?= e($_SESSION['role']) ?></p>
<?php else: ?>
    <h1>Login</h1><?php if ($error): ?><p><?= e($error) ?></p><?php endif; ?>
    <form method="post">
        <label>Username <input type="email" name="email" required></label>
        <label>Password <input type="password" name="password" required></label>
        <button type="submit">Login</button>
    </form>
<?php endif; ?>
</body></html>
```

Marker note

A fixed exercise password is retained only to match the historical paper. Real applications must store password hashes and use `password_verify()`.